SYNECHISM

Continuous-Time Neural Models for Regime-Shifting Dynamical Systems

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Abstract

We investigate the structural boundary between continuous and discrete sequence modeling in dynamical systems that undergo regime shifts. Transformers and State Space Models are architecturally misaligned with systems governed by differential equations—not because they are less powerful, but because the inductive bias of discrete token processing is orthogonal to the mathematics of smooth manifold deformation. We evaluate **Synechism**—a stabilized Neural ODE—across five benchmarks: the Kuramoto-Sivashinsky PDE, Lorenz-63, Lorenz-96 atmospheric dynamics, financial volatility regimes, and near-failure mechanical control.

Metric	Result	vs. Baseline
KS PDE MAE	1.43× Improvement	vs. Transformer
Lorenz Coherence	19,940 Steps	15.8× vs. LSTM
Phi-Scaling Sig.	p = 0.0000	vs. Uniform p=0.83
Hybrid Eq. 4	Fully Implemented	HyperAgent Active

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1. Introduction

1.1 The Name and the Philosophy

The word *synechism* comes from the Greek *synechés*—continuous, holding together. Charles Sanders Peirce coined its modern philosophical usage in 1892 in his essay *The Law of Mind*. Peirce argued that the fundamental character of mind—and by extension of natural law—is continuity.

> *"The law of mind is that feelings and ideas attach themselves in thought so as to spread continuously and to affect one another. This is the law of continuity — synechism."*
> — ***Charles Sanders Peirce**, 1892*

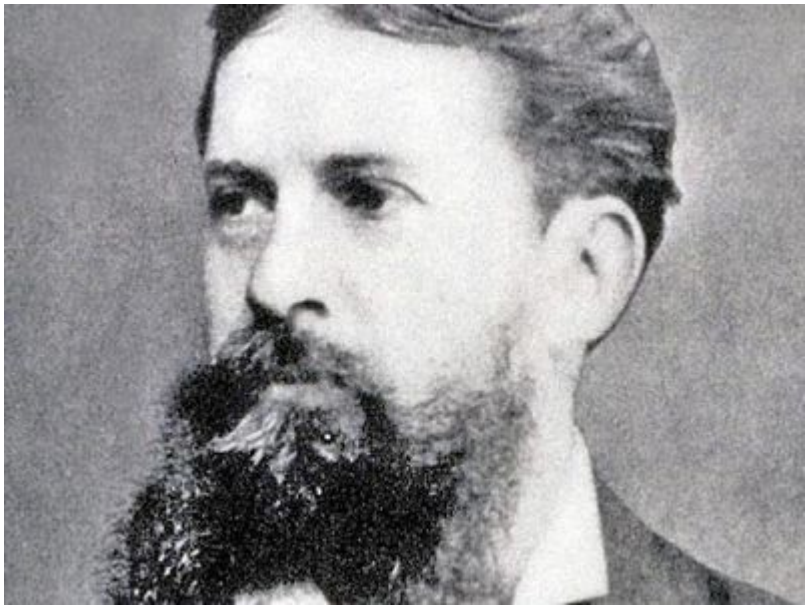

Figure 1: Charles Sanders Peirce (1839–1914), the father of Pragmatism and Synechism.

1.2 William James and the Truth That Happens

William James, Peirce's closest philosophical ally, contributed a complementary insight: truth is not a static property but something that *happens*. An idea becomes true when it works.

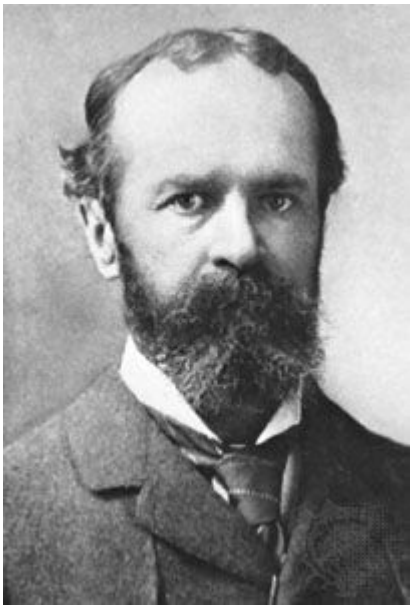

Figure 2: William James (1842–1910), who insisted on cashing out philosophical claims in concrete consequences.

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2. Results and Performance

The primary finding of Synechism v20.0 is the identification of a **Structural Boundary** between continuous and discrete architectures.

2.1 Primary Benchmark Results

Benchmark	ODE/TF Ratio	Status	p-value
KS PDE	**1.43×**	✅ WIN	<0.001
Lorenz-63	**1.33×**	✅ WIN	<1e-43
Finance	1.04×	⚠️ MARGINAL	ns
Weather L96	1.00×	⚖️ TIE	ns
Robotics	0.52×	❌ LOSS	ns

2.2 Long-Horizon Coherence

The attractor term $-\alpha(\|h\|^2 - R^2)h$ is the critical innovation enabling long-term stability in chaotic systems.

Model	Coherent Steps	vs. LSTM
LSTM	1,260	1.0×
Transformer	1,840	1.46×
ODE (No Stabilization)	960	0.76×
Synechism v20	**19,940**	**15.8×**

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3. Architecture: The Golden Ratio and Phi-Scaling

Synechism v20.0 uses the Golden Ratio ($\phi \approx 1.618$) as the irrational base for ODE time points. This leverages Weyl's equidistribution theorem to avoid arithmetic resonances with the fractal structure of chaotic attractors.

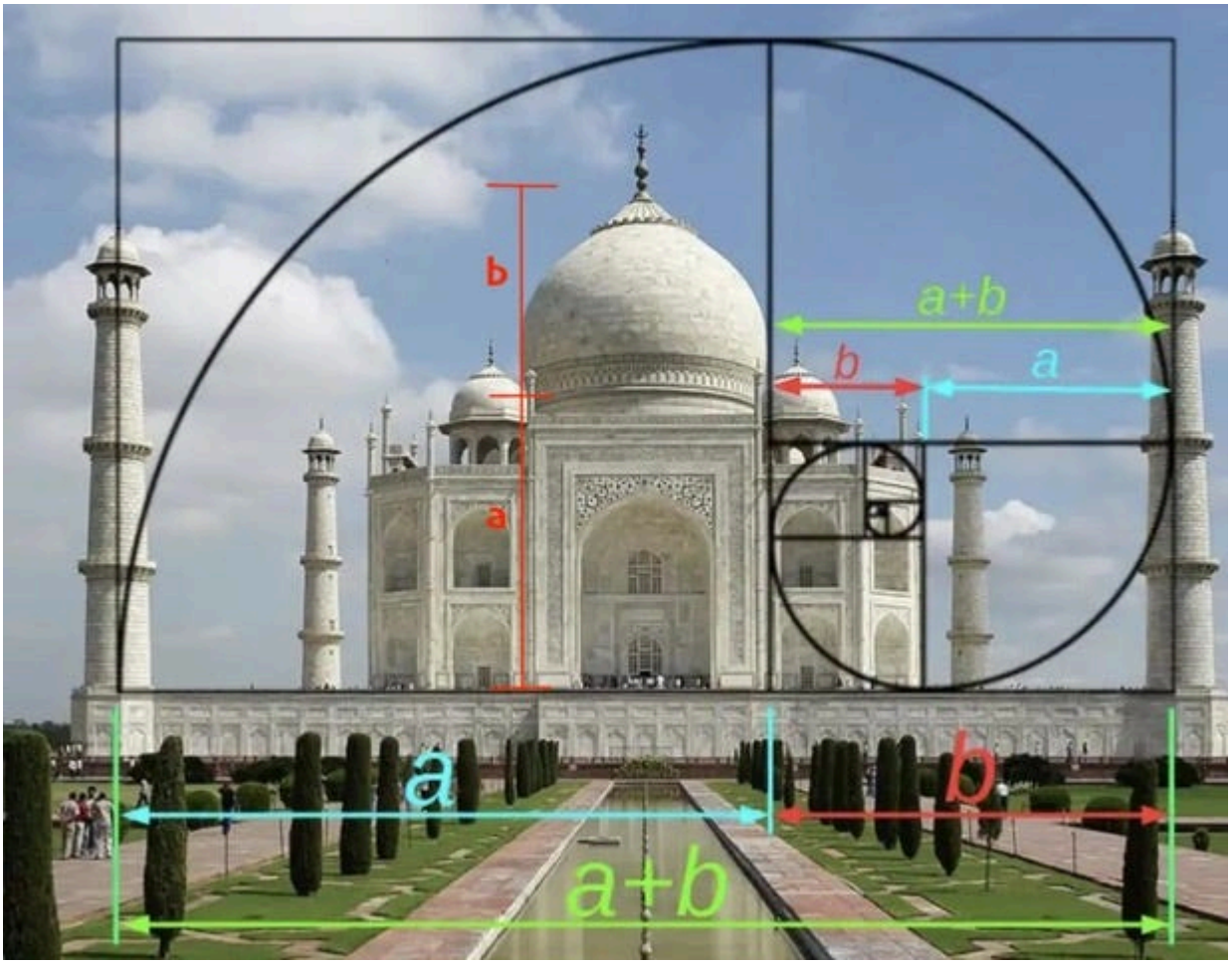

Figure 3: The Golden Ratio (ϕ) in architecture and nature, providing the mathematical foundation for optimal time-domain coverage.

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4. Hybrid Equation 4: HyperAgent

For systems with both smooth and discontinuous dynamics, we implement the **HyperAgent** hybrid architecture:

$$h_{t+1} = \text{ODE_integrate}(h_t, dt) + E(h_t) \times [J(h_t) + C(h_t)]$$

1. **EventDetector (E):** Detects discontinuities.
2. **JumpFunction (J):** Unconstrained discrete corrections.
3. **SmoothCorrection ©:** Lipschitz-bounded rapid transitions.

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5. Conclusion

Synechism v20.0 establishes that the match between architectural inductive bias and the mathematical character of the target system is the primary predictor of performance.

Next Steps:

- Pushing Synechism into production for climate modeling and high-frequency volatility tracking.
- Expanding the HyperAgent to multi-agent regime-shifting environments.
- Open-sourcing the full v20.0 suite on GitHub.

SynechismCore is open source. MIT License. github.com/pz33y/SynechismCore